ModelIt!

Teacher's Guide

How Does Your Phone Fit 10,000 Songs?

Modeling Digital Signal Processing, Data Compression, and Recommendation Algorithms

Quick Reference

Grade Level	Grade 10L1
NGSS Standard	HS-PS4-5, HS-ETS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS4-5, HS-ETS1-1

Communicate technical information about how some technological devices use the principles of wave behavior and wave interactions with matter to transmit and capture information and energy. Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students construct explanations of how wave behavior enables digital audio capture and design solutions that balance quality, storage, and user experience.
Disciplinary Core Idea	**PS4.C Information Technologies and Instrumentation / ETS1.A Defining and Delimiting Engineering Problems** — Technological devices use wave principles to capture, process, and transmit audio information. Engineering problems require specifying criteria and constraints.
Crosscutting Concept	**Systems and System Models** — Students model the music streaming system as interconnected subsystems — signal capture, compression, storage, and recommendation — that must be optimized together.

CAST Testing Alignment

MULTIPLE CHOICE:

An uncompressed WAV file of a 3-minute song is 30MB. After MP3 compression at 128kbps, the file is 2.8MB. What percentage of the original data was removed, and why don't most listeners notice?

CONSTRUCTED RESPONSE:

Using your model, explain how a music streaming service can deliver audio that sounds nearly identical to the original recording while using only 10% of the data. Connect your explanation to wave behavior, human hearing perception, and compression algorithm design.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The ability to carry thousands of songs in your pocket represents one of the most remarkable applications of wave physics and computational mathematics in history. Every song on your phone began as vibrations in air, was converted to electrical signals, digitized into numbers, compressed by algorithms that exploit human hearing limits, and stored as magnetic patterns on a tiny chip. Understanding this chain means understanding both the physics of waves and the engineering of information.

Sound as Waves

Sound is a longitudinal pressure wave traveling through air at about 343 m/s. The wave's frequency (20 Hz to 20,000 Hz for human hearing) determines pitch, and its amplitude determines loudness. When a microphone captures sound, it converts these pressure variations into an analog electrical signal that mirrors the original wave pattern.

Analog-to-Digital Conversion

To store sound digitally, the analog signal must be sampled — measured at regular intervals — and quantized — rounded to the nearest digital value. CD quality uses 44,100 samples per second (Nyquist theorem requires 2x the highest frequency) at 16-bit depth (65,536 possible values per sample). This produces about 10MB of raw data per minute of stereo music.

Psychoacoustic Compression

MP3, AAC, and Ogg Vorbis compression algorithms exploit the limits of human hearing through a technique called perceptual coding. Sounds below the hearing threshold are removed. When a loud sound masks a quiet sound (simultaneous masking), the quiet sound is removed. Sounds that occur just before or after a loud sound are also removed (temporal masking). The result: 90% data reduction with minimal perceived quality loss.

Machine Learning Recommendations

Modern recommendation systems use collaborative filtering (users who liked X also liked Y), content analysis (analyzing the audio features of songs — tempo, key, energy, danceability), and deep learning (neural networks that find complex patterns in billions of listening sessions). Spotify's 'Discover Weekly' playlist analyzes over 100 data points per song per listener to generate 30 personalized recommendations each Monday.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify audio signal quality, data compression rate, algorithm accuracy, user engagement, and recommendation match as the key components of the digital music system.

E **Establish Relationships**
Students determine that audio quality and compression rate have an inverse relationship, and that algorithm accuracy improves through a feedback loop with user engagement — more data means better predictions.

V **Visualize & Model**
Students build a computational model showing how compression settings affect storage capacity and audio fidelity, and how user data volume drives recommendation performance.

E **Evaluate Outcomes**
Students run high quality, maximum compression, and new-vs-veteran user scenarios to discover the perception threshold for compression and the data requirement for accurate recommendations.

R **Revise & Extend**
Students add bit depth, network bandwidth, or listening context to model more realistic streaming system dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Your phone has 10,000 songs on it. Each song started as vibrations in a recording studio. How did AIR VIBRATIONS become DATA in your pocket?"*

Watch For

Show file size comparison: one uncompressed WAV song (50MB) vs. 10,000 compressed songs on a 64GB phone. How is that possible?

Slide 2: Learning Objectives

3 min

Say: *"Today we're tracing the journey of a sound wave from a singer's mouth to your earbuds — and every transformation along the way."*

Watch For

Have students read objectives. Pre-teach 'data compression' and 'sampling rate' with audio examples.

Slide 3: Big Question

3 min

Say: *"If one song takes 50MB uncompressed, 10,000 songs would need 500GB. Your phone has 64GB. Something is being removed. What?"*

Watch For

Students calculate: How much data must be removed? What could possibly be expendable in a song?

Slide 4: LEVER Framework

2 min

Say: *"We're modeling the entire chain — from sound wave to compressed file to recommended playlist."*

Watch For

Preview LEVER steps. Emphasize this connects wave physics to information technology.

Slide 5: Activity 1: Components

8 min

Say: *"What can a streaming service CONTROL? What outcomes emerge from those choices?"*

Watch For

Guide sorting. Audio quality and compression rate are design choices; engagement and match are outcomes.

Slide 6: Activity 2: Connections

8 min

Say: *"More compression means smaller files but worse sound. Where's the sweet spot where your ears can't tell the difference?"*

Watch For

Play audio samples at different compression levels. Can students identify which is compressed?

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's find the threshold — the exact compression level where your ears stop noticing the difference."*

Watch For

Students predict the perception threshold. Run simulations. Then test new-user vs. veteran recommendation accuracy.

Slide 8: Discoveries

5 min

Say: *"The algorithm removes sounds you literally CANNOT hear. And the recommendation engine gets better every time you press play or skip."*

Watch For

Connect psychoacoustic masking to wave physics. Discuss the feedback loop driving algorithm improvement.

Slide 9: STEM Challenge

12 min

Say: *"Build a music app for students with cheap phones and bad wifi. Make it sound good, fit a lot of songs, and recommend what they'll love."*

Watch For

> Groups design streaming systems balancing quality, storage, and recommendation. Present solutions.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Audio Signal Quality, Data Compression Rate
- **INTERNAL:** Algorithm Accuracy, User Engagement, Recommendation Match

Relationships

- **Audio Signal Quality to Data Compression Rate:** INVERSE — Higher audio quality means larger raw files that require more aggressive compression to fit in limited storage. Engineers must balance capturing high-fidelity sound against the need to reduce file size for practical storage and streaming.
- **Data Compression Rate to User Engagement:** COMPLEX (+/-) — Moderate compression maintains audio quality that keeps users engaged. But excessive compression degrades quality enough that users disengage — there is an optimal zone where files are small enough to store efficiently but good enough to enjoy.
- **User Engagement to Algorithm Accuracy:** POSITIVE (+) — More user engagement generates more listening data — play counts, skip rates, completion rates, time-of-day patterns. This data trains the recommendation algorithm, improving its accuracy. The more you listen, the better it knows you.

Key Discoveries

- Data compression exploits human perception limits — MP3 removes frequencies most people can't hear and sounds masked by louder sounds, reducing file size by 90% with minimal perceived quality loss
- There's a critical compression threshold beyond which quality drops noticeably — the algorithm must balance file size against the limits of human hearing perception
- Recommendation algorithms improve through a feedback loop: more listening generates more data, which improves predictions, which increases engagement, which generates more data
- The entire modern music industry depends on the same wave physics principles that govern how sound travels through air — digitization is simply converting those waves to numbers

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Compressed music sounds terrible compared to the original"	Modern lossy compression at 256kbps or higher is virtually indistinguishable from the original in controlled listening tests. The compression algorithm specifically removes sounds that fall below human perception thresholds — frequencies you can't hear and sounds masked by louder sounds. Most people cannot identify the compressed version in a blind A/B test at standard quality settings.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"More data always means better recommendations"	While more data generally improves recommendations, there are diminishing returns and quality matters more than quantity. An algorithm with 100 hours of diverse listening data often outperforms one with 1,000 hours of repetitive data. Additionally, more data collection raises privacy concerns — the question isn't just CAN we collect more data, but SHOULD we.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Digital music perfectly captures the original sound"	Digitization is a sampling process that captures snapshots of the sound wave at regular intervals. Between samples, the original analog information is lost and must be interpolated. At CD quality (44.1kHz), the Nyquist theorem guarantees frequencies up to 22.05kHz are captured accurately — sufficient for human hearing (20Hz-20kHz). But no digital format captures every infinitesimal detail of the original analog wave.